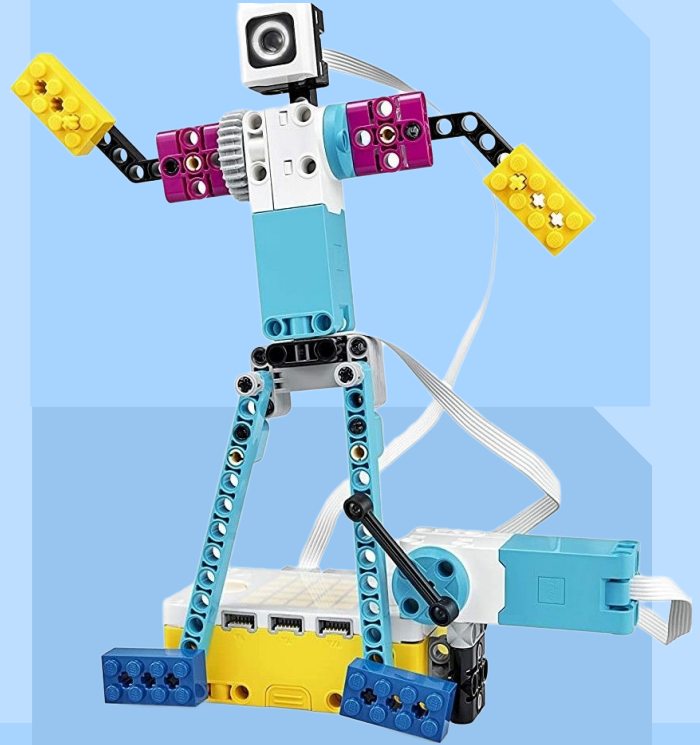


LESSON 4

Intro to Spike Prime



1. Stored Energy

[Elastic Potential Energy]



The Question

"What happens when you stretch a rubber band?"

Answer:

The rubber band stores energy.

When Released

- Stored energy becomes **motion**.
- Motion **launches** the projectile.

2. Levers

Vocabulary

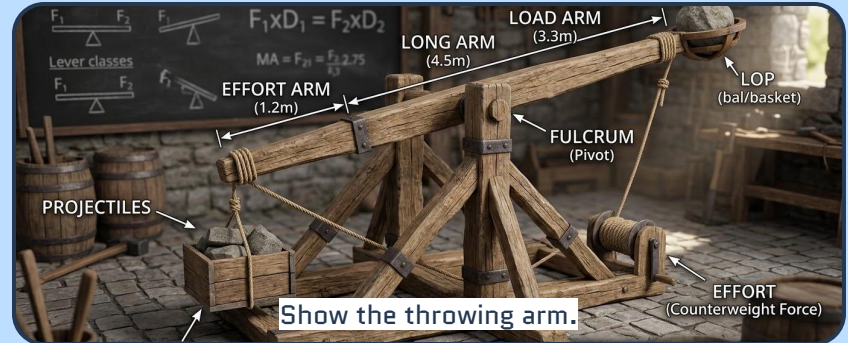
- **Fulcrum:** pivot point
- **Effort:** where force is applied
- **Load:** the projectile

Ask:

"What happens if we make the arm longer?"

Possible Answers:

- Projectile travels farther
- More speed
- Harder to control

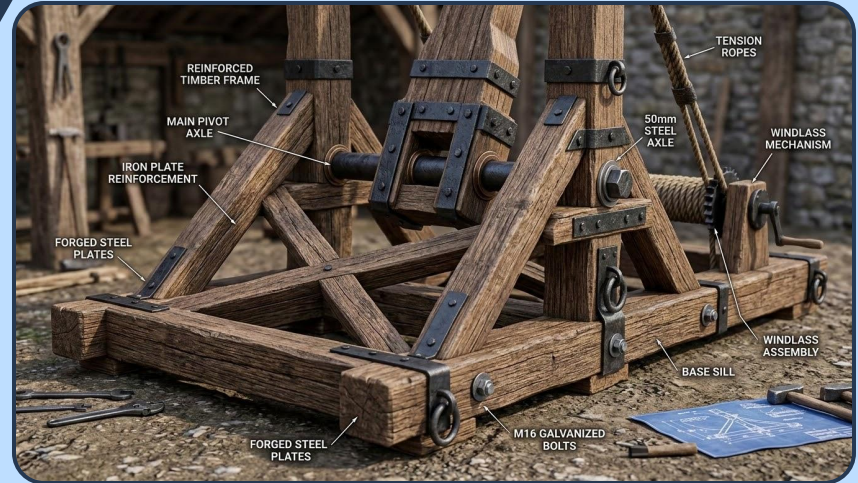


3. Stability

Ask: "What happens if the catapult tips over?"

Engineers Need:

- Wide base
- Strong frame
- Reinforcement

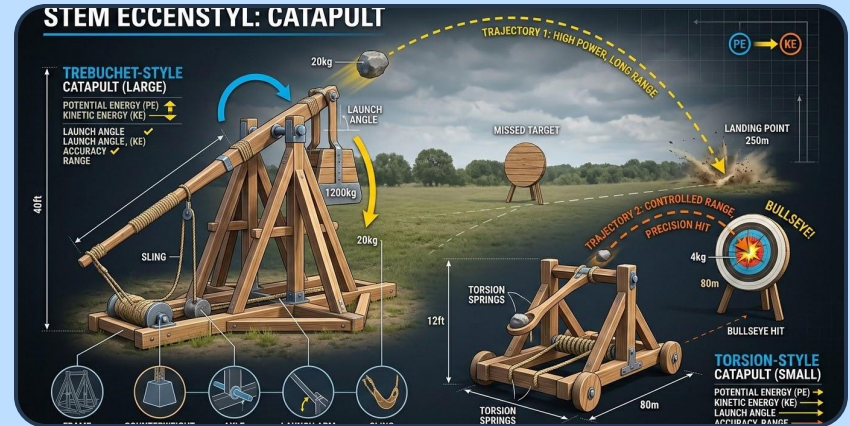


Accuracy vs Distance

Ask: "Is the strongest catapult always the best?"

Not necessarily.

A catapult that launches 20 feet but misses the target may lose to one that launches 10 feet accurately.



Challenge #1: Distance Competition

Objective: Launch the projectile the farthest.

Rules:

- Same projectile for everyone
- Launch from same line
- Best of 3 attempts



Award:



Longest Launch

Challenge #2: Target Accuracy

Students will set up cups, buckets, or taped targets. Adjust your catapult to hit a target that is 6 inches away from your catapult.

Point System:

- Each team gets 3 shots.
- Highest score wins.

Award:



Best Accuracy